

## CARGO SPACE

**Cargo space**, as the name implies, is the space available for cargo to be carried on board different types of ship. It can be expressed as the grain capacity or bale capacity of the ship and can be found in the holds, in the 'tween decks, in the tanks and deep tanks, in refrigerated compartments and cooling chambers and in case of certain cargoes such as timber or cargoes carried in containers, even on deck .

**Bale capacity** is the cubic capacity of any space available for cargo such as bales of wool measured from the ceiling of the hold to the underside of the deck beams (the depth), between the inside of the cargo battens (the breadth) and between the inside of the bulkheads or sparring where fitted (the length). It is measured in cubic metres or cubic feet.

**Grain capacity** is the cubic capacity of any space available for cargo such as grain which fills the hold entirely. It is the total capacity in the hold with an allowance for the volume occupied by frames and beams. It is measured in cubic metres or cubic feet. It is greater than the bale capacity of the vessel.

**Broken stowage** is the space between packages not taken up by the cargo, the space which remains unfilled for to a variety of reasons such as the shape of the hold, the type of cargo, special kinds of packing, irregularly-sized items of machinery etc. It is expressed as percentage which is usually greater when large cases have to be stowed in the hold.

Cargo space on board container vessels is usually measured and expressed in TEU which stands for Twenty-Foot Equivalent Unit.

**Cargo space on board Ro-Ro** ships is expressed in the length of lanes and is measured in metres. It can also be expressed as bale capacity and measured in cubic metres or cubic feet.

**Cargo space on board Con-Ro** vessels is expressed both in TEU for the number of containers the vessel can carry and in the number of cars it can take on car decks so the length of lanes is also given.

**Cargo space** can be filled with different types of cargo. It can usually be divided into space for general cargo which is packed and bulk cargo, both liquid and dry, which is

loose. Different liquids and chemicals can evaporate from their receptacles and tanks and in such cases the unfilled space is called ullage.

**General cargo** can be divided into containerised cargo, non-containerised cargo and refrigerated cargo. General cargo may cause many stowage problems because the goods can be packed in different cases, bags, boxes, bundles, crates and drums and some pieces of machinery or heavy lifts can be loaded without any packaging at all. In such cases the broken stowage is very high.

**Containerised cargo** prevails nowadays and causes least problems in sea transport because containers are of standard shape and dimensions. **Perishable cargo** is loaded into refrigerated containers or refrigerated holds and carried in reefer ships, which are specially built for that purpose. Perishable cargo includes meat, fish, dairy produce and fruit. **Cars** can be carried on board the PCC which stands for Pure Car Carriers and cargo space in those ships is expressed in the length of lanes and the number of cars they can carry.

**Bulk cargo** can be divided into liquid cargo and dry cargo. Liquid cargo such as crude oil and its products can be carried in tankers. Dry bulk cargo is usually carried by bulkers or bulk cargo vessels. OBO vessels carry both types of bulk cargo. OBO stands for Oil/Bulk/Ore.

**LPG and LNG** vessels carry liquefied petroleum gas and liquefied natural gas in specially constructed steel spheres under pressure and at low temperatures. Chemicals such as molten sulphur are carried in special tanks by chemical carriers.